

# Numerical Experiments

## Geometry-Dependent Modal Requirements of Finite-Area Interface Representations in Affine-DDPNM

Standalone working version of §6–§10 of the full paper

**Current numerical scope.** Only completed benchmark results are assigned numerical values below. The monolithic FEM reference and the theoretical broken-pressure C-DD reference are kept distinct throughout. Experiments that remain to be run are identified explicitly in §10.3; in particular, no full-C-DD Schur-energy values or  $\mathcal{W}_{0v}$  accuracy values are inferred from the available FEM-relative data.

## 6 Numerical methodology

### 6.1 Geometries and discretisations

Three three-dimensional test geometries are used. All are unit-cube packings of solid spheres with Stokes flow in the remaining fluid domain, partitioned into  $N_p$  pore subdomains with  $m$  planar internal interfaces.

- (i) **Uniform-27:** a regular  $3 \times 3 \times 3$  lattice of 27 equal spheres (radius 0.105) centred at  $\{0.20, 0.50, 0.80\}^3$ . The partition is the regular  $4 \times 4 \times 4$  grid cut, giving  $N_p = 64$  pores and  $m = 144$  interfaces (mean  $2m/N_p = 4.5$  ports per pore).
- (ii) **Random-27:** a frozen random packing of 27 spheres (seed 20260804, 18 wall-clipped and 9 interior spheres); the partition is the Voronoi tessellation of the sphere centres, giving  $N_p = 27$  pores and  $m = 114$  interfaces (mean 8.44 ports per pore).
- (iii) **Heterogeneous-100:** a synthetic frozen random packing of 100 spheres (project-internal label “Real-100”, seed 20260805) with porosity 0.856; the partition is the regular  $4 \times 4 \times 4$  grid cut, giving  $N_p = 64$  pores and  $m = 144$  interfaces (mean 4.5 ports per pore).

The solid-phase packings are shown schematically in Fig. 1 (the finite-element meshes themselves are not rendered; the sphere positions and radii are those used to build the meshes). Mesh statistics are summarised in Table 1, including the objective variability descriptors suggested by the review of the experimental design:

$$\text{CV}(A_\gamma) = \frac{\text{std}(A_\gamma)}{\text{mean}(A_\gamma)}, \quad \text{CV}(V_i) = \frac{\text{std}(V_i)}{\text{mean}(V_i)}, \quad \text{CV}(a_\gamma/b_\gamma) = \frac{\text{std}(a_\gamma/b_\gamma)}{\text{mean}(a_\gamma/b_\gamma)}, \quad (1)$$

where  $A_\gamma$  is the area of interface  $\gamma$ ,  $V_i$  the volume of pore  $i$ , and  $a_\gamma/b_\gamma \geq 1$  the in-plane aspect ratio of the interface patch, estimated as the square root of the ratio of the two largest principal-component eigenvalues of the patch vertices. These are *reported* as descriptive quantities; no claim is made that a single scalar controls the error.

The three configurations comprise a deliberately regular lattice (Uniform-27), a deliberately irregular Voronoi-partitioned case (Random-27), and a larger synthetic heterogeneous packing (Heterogeneous-100). The partition-level descriptors in Table 1 should not be interpreted as a scalar measure of “geometric complexity”: Heterogeneous-100 uses a regular grid partition and therefore has small  $\text{CV}(V_i)$ ,  $\text{CV}(A_\gamma)$  and  $\text{CV}(a_\gamma/b_\gamma)$  despite its heterogeneous solid arrangement.

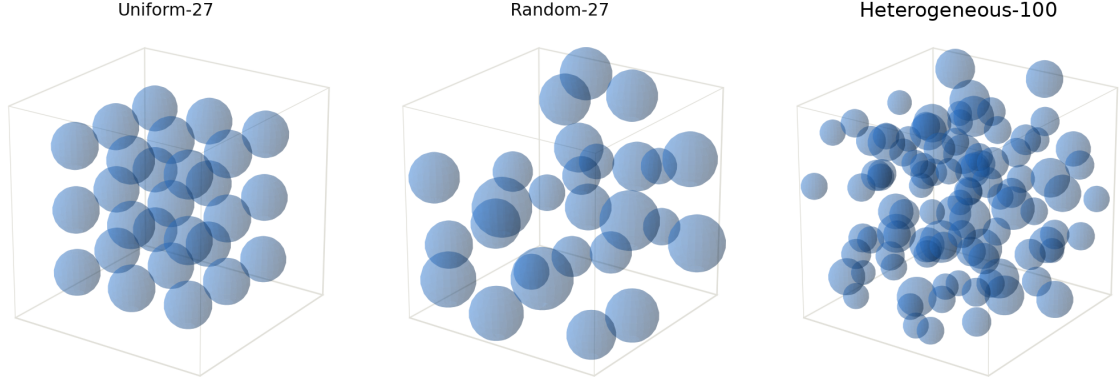


Figure 1: Schematic of the three solid-phase packings used to build the test geometries (sphere positions and radii as used by the mesh generator; finite-element meshes are not shown): (a) Uniform-27 regular  $3 \times 3 \times 3$  lattice; (b) Random-27 frozen packing of 27 spheres; (c) Heterogeneous-100 frozen packing of 100 spheres.

Table 1: Geometry and discretisation summary. FEM DOF counts are for the monolithic Taylor–Hood pair  $[P_2]^3-P_1$  (velocity DOFs =  $3 \times$  number of P2 nodes; pressure DOFs = number of P1 nodes); they match the mixed-unknown counts reported by the solver.

| Geometry          | spheres | $N_p$ | $m$ | cells  | vel.    | pres. | mixed   | $2m/N_p$ | remarks                  |
|-------------------|---------|-------|-----|--------|---------|-------|---------|----------|--------------------------|
| Uniform-27        | 27      | 64    | 144 | 12 203 | 60 936  | 3 019 | 63 955  | 4.5      | regular lattice          |
| Random-27         | 27      | 27    | 114 | 15 249 | 72 471  | 3 483 | 75 954  | 8.44     | Voronoi, irregular       |
| Heterogeneous-100 | 100     | 64    | 144 | 32 611 | 159 186 | 7 655 | 166 841 | 4.5      | grid cut, porosity 0.856 |

| Geometry          | CV ( $V_i$ ) | CV ( $A_\gamma$ ) | CV ( $a_\gamma/b_\gamma$ ) | mean $a_\gamma/b_\gamma$ |
|-------------------|--------------|-------------------|----------------------------|--------------------------|
| Uniform-27        | 0.321        | 0.203             | 0.158                      | 1.39                     |
| Random-27         | 0.432        | 0.783             | 1.479                      | 2.48                     |
| Heterogeneous-100 | 0.057        | 0.116             | 0.078                      | 1.11                     |

## 6.2 Reference systems and error metrics

Two reference systems must be kept distinct (see also the theory document):

- (i) **Physical/numerical reference:** the monolithic Taylor–Hood finite-element solution  $\mathbf{u}_h^{\text{FEM}}, p_h^{\text{FEM}}$  on the *same* mesh, with continuous pressure across pores. All reported accuracy metrics are relative to this reference. The theoretical section does not establish equivalence between this system and the broken-pressure C-DD reference; the error relative to the FEM solution is therefore reported as an *observed* accuracy measure, and no rate statement is made about it from the theory.
- (ii) **Theoretical Galerkin reference:** the broken-pressure C-DD trace solution  $\xi_{\text{CDD}}$  of the theory sections, and the Schur-energy seminorm  $\|\cdot\|_{\mathbf{S}_{uu}}$  it induces. The theory-level quantities ( $\|\xi_{\text{CDD}} - \xi_X\|_{\mathbf{S}_{uu}}$ , the Pythagorean decompositions) were *not* computed in the runs archived so far (the full C-DD solve was not executed); they are discussed in §7.3 as planned verification, and no numerical values are reported for them.

Table 2: The four modal interface spaces and their global unknown budgets on the three geometries ( $m = \text{number of interfaces: } 144/114/144$ ). The row marked  $\dagger$  was *not computed* in the runs archived so far.

| Method              | space/interface    | modes/iface | global unknowns |           |                   |
|---------------------|--------------------|-------------|-----------------|-----------|-------------------|
|                     |                    |             | Uniform-27      | Random-27 | Heterogeneous-100 |
| Classic             | $\mathcal{W}_{0n}$ | 1           | 144             | 114       | 144               |
| P0-vector $\dagger$ | $\mathcal{W}_{0v}$ | 3           | 432             | 342       | 432               |
| NormalLinear        | $\mathcal{W}_{1n}$ | 3           | 432             | 342       | 432               |
| Affine              | $\mathcal{W}_{1v}$ | 9           | 1296            | 1026      | 1296              |

Per-interface contents:  $\mathcal{W}_{0n} = \{1\mathbf{n}\}$ ;  $\mathcal{W}_{0v} = \{1\mathbf{n}, 1\mathbf{t}_1, 1\mathbf{t}_2\}$ ;  $\mathcal{W}_{1n} = \{1\mathbf{n}, s\mathbf{n}, t\mathbf{n}\}$ ;  $\mathcal{W}_{1v} = \{1, s, t\} \otimes \{\mathbf{n}, \mathbf{t}_1, \mathbf{t}_2\}$ .  $\dagger$

Constant-vector (P0-vector) space: basis implemented as part of the four-space framework but not run in the archived benchmarks; included for the equal-budget comparison in §8.4.

For each reduced space  $X \in \{\mathcal{W}_{0n}, \mathcal{W}_{0v}, \mathcal{W}_{1n}, \mathcal{W}_{1v}\}$  we report, uniformly:

$$e_u^{L^2} = \frac{\|\mathbf{u}_X - \mathbf{u}_{\text{FEM}}\|_{L^2(\Omega)}}{\|\mathbf{u}_{\text{FEM}}\|_{L^2(\Omega)}}, \quad (2)$$

$$e_u^{H_{\text{br}}^1} = \frac{\left(\sum_i \|\nabla(\mathbf{u}_X - \mathbf{u}_{\text{FEM}})\|_{L^2(\Omega_i)}^2\right)^{1/2}}{\left(\sum_i \|\nabla \mathbf{u}_{\text{FEM}}\|_{L^2(\Omega_i)}^2\right)^{1/2}}, \quad (3)$$

$$e_p^{L^2} = \frac{\|p_X - p_{\text{FEM}}\|_{L^2(\Omega)}}{\|p_{\text{FEM}}\|_{L^2(\Omega)}}, \quad (4)$$

$$e_Q = \frac{|Q_X - Q_{\text{FEM}}|}{|Q_{\text{FEM}}|}, \quad Q = \text{outlet volume flux.} \quad (5)$$

All error norms are evaluated cell-wise on the common mesh (broken domain), with degree-6 quadrature. The pressure gauge is fixed by Dirichlet pressure conditions at inlet ( $p = 1$ ) and outlet ( $p = 0$ ) in both the FEM and the reduced systems; the raw relative  $L^2$  pressure error (4) is reported without any post-hoc mean alignment. Because the pressure level is fixed physically by the inlet/outlet conditions, this is the pressure metric used consistently below. The theory gives no monotonicity of (4) along the nested chains, so occasional non-monotone pressure errors are not a discrepancy.

### 6.3 Four modal interface spaces

The numerical programme is organised around the four trace subspaces of the theory section, whose per-interface contents are

$$\mathcal{W}_{0n} : \{1 \cdot \mathbf{n}\}, \quad \mathcal{W}_{0v} : \{1 \cdot \mathbf{n}, 1 \cdot \mathbf{t}_1, 1 \cdot \mathbf{t}_2\}, \quad (6)$$

$$\mathcal{W}_{1n} : \{1 \cdot \mathbf{n}, s \cdot \mathbf{n}, t \cdot \mathbf{n}\}, \quad (7)$$

$$\mathcal{W}_{1v} : \{1, s, t\} \otimes \{\mathbf{n}, \mathbf{t}_1, \mathbf{t}_2\}, \quad (8)$$

where  $(s, t)$  are local planar coordinates on the (representative) interface and  $\{\mathbf{n}, \mathbf{t}_1, \mathbf{t}_2\}$  a fixed orthonormal frame. Table 2 lists the unknown budgets. The two three-mode spaces satisfy  $\dim \mathcal{W}_{0v} = \dim \mathcal{W}_{1n} = 3m$ , which makes them an *equal-budget* pair: the same number of interface unknowns spent on directional content ( $\mathcal{W}_{0v}$ ) versus on first-order spatial content ( $\mathcal{W}_{1n}$ ). Of the two paths  $\mathcal{W}_{0n} \subseteq \mathcal{W}_{0v} \subseteq \mathcal{W}_{1v}$  and  $\mathcal{W}_{0n} \subseteq \mathcal{W}_{1n} \subseteq \mathcal{W}_{1v}$ , only the second was executable with the implemented bases at the time of the archived runs;  $\mathcal{W}_{0v}$  (constant tangential modes) is listed as *not computed* in this study.

## 6.4 Timing and solver protocol

All archived runs used the same serial pipeline on one Windows 11 x86-64 workstation, with viscosity 1 and inlet/outlet pressure 1/0. The exact CPU/RAM and software-version metadata are not recorded in this working draft and must be added for the final performance benchmark. Timings distinguish

$$T_{\text{offline}} = \text{local factorisations and primitive response construction}, \quad (9)$$

$$T_{\text{online}} = \text{global reduced assembly} + \text{Schur solve} + \text{local reconstruction}. \quad (10)$$

$T_{\text{first}} = T_{\text{offline}} + T_{\text{online}}$  is the wall time of a first reduced solve. Common mesh generation and post-processing validation are excluded. The archived timings are not yet a fully uniform performance benchmark: for Uniform-27 and Random-27 the FEM entry is assembly+solve, whereas the Heterogeneous-100 archive contains FEM solve time only. Consequently Table 5 reports absolute timing scopes explicitly. First-solve speedups are quoted only where the FEM and reduced timings are directly comparable; online-only ratios for Heterogeneous-100 are labelled as such and are not interpreted as single-solve wall-clock speedups.

## 7 Verification of the reference implementation

### 7.1 Exact-FE-Schur: Schur elimination code validation

On Random-27 the benchmark additionally solves the exact Schur complement of the monolithic system on the full FE interface trace (21 970 trace DOFs, 465.9 s). This serves as a code-level validation of the Schur-elimination machinery, *not* as the theoretical C-DD reference: the two systems share the assembled monolithic operator and must agree to round-off. The measured relative difference between the exact-FE-Schur solution and the monolithic solve is

$$\frac{\|\mathbf{u}_{\text{Schur}} - \mathbf{u}_{\text{direct}}\|}{\|\mathbf{u}_{\text{direct}}\|} = 1.05 \times 10^{-12}, \quad (11)$$

with Schur symmetry defect  $3.2 \times 10^{-15}$ , Schur relative residual  $5.75 \times 10^{-14}$  and global relative residual  $1.55 \times 10^{-15}$ . This validates the full-trace Schur elimination and the reference/error-evaluation pipeline. The reduced systems are checked separately through their Schur residuals, symmetry and conservation diagnostics below; the Exact-FE-Schur comparison is not, by itself, a validation of the affine modal-basis assembly.

### 7.2 Conservation and algebraic diagnostics

Table 3 collects the algebraic diagnostics recorded by the runs. Local discrete incompressibility gives each pore balance, while the retained constant-normal Schur equation enforces cancellation of the internal interface fluxes; together these yield the global mass balance under the assumptions of the conservation theorem. The residuals shown are the maximum absolute modal-moment residual entries of the assembled Schur systems, at round-off level in all recorded cases. The Schur matrices are exactly symmetric by construction (symmetry defect zero for the reduced systems). On Random-27 the recorded minimum Schur eigenvalue drops from  $2.1 \times 10^{-8}$  (Classic) to  $1.26 \times 10^{-12}$  for both NormalLinear and Affine. This indicates a much smaller minimum spectral scale in the chosen coefficient normalization, but it is not by itself a condition-number estimate because  $\lambda_{\text{max}}(\mathbf{S})$  was not archived. The matrices were successfully factorised as SPD in these runs.

### 7.3 Theory-level verification: not yet computed

The theory sections derive the best-approximation property in the  $\hat{\mathbf{S}}^{uu}$ -seminorm and the exact Pythagorean decompositions along the two nested paths, with the reduction errors  $\Delta_T =$

Table 3: Numerical verification diagnostics as recorded by the archived runs (“–” = not recorded by that benchmark). “max  $r_{\text{mom}}$ ” is the maximum absolute modal-moment residual of the assembled reduced system; the mass-balance row is the inlet/outlet flux sum of the random benchmark.

| Geometry          | Method   | max $ r_{\text{mom}} $ | Schur sym. defect | $\lambda_{\min}(\mathbf{S})$ | in/out flux sum      |
|-------------------|----------|------------------------|-------------------|------------------------------|----------------------|
| Uniform-27        | all      | –                      | –                 | $1.67 \times 10^{-4}\dagger$ | –                    |
| Random-27         | $W_{0n}$ | $2.17 \times 10^{-16}$ | 0                 | $2.11 \times 10^{-8}$        | $-5 \times 10^{-16}$ |
| Random-27         | $W_{1n}$ | $3.55 \times 10^{-16}$ | 0                 | $1.26 \times 10^{-12}$       | $-2 \times 10^{-16}$ |
| Random-27         | $W_{1v}$ | $4.92 \times 10^{-16}$ | 0                 | $1.26 \times 10^{-12}$       | $-4 \times 10^{-16}$ |
| Heterogeneous-100 | $W_{0n}$ | $6.36 \times 10^{-17}$ | 0                 | –                            | –                    |
| Heterogeneous-100 | $W_{1n}$ | $1.05 \times 10^{-16}$ | 0                 | –                            | –                    |
| Heterogeneous-100 | $W_{1v}$ | $1.45 \times 10^{-16}$ | 0                 | –                            | –                    |

Recorded by the base-project verification run (`ddpnm_3d_uniform_spheres`, same geometry, Classic system); the affine benchmark did not record the eigenvalue. The monolithic FEM reference solves reached relative linear residuals  $1.73 \times 10^{-12}$  (Uniform-27) and  $2.92 \times 10^{-12}$  (Random-27), with relative mass imbalance  $2.54 \times 10^{-14}$  (Random-27).

Table 4: Complete accuracy results relative to the monolithic Taylor–Hood FEM on the same mesh.  $e_u^{L^2}$ ,  $e_u^{H^1_{\text{br}}}$ ,  $e_p^{L^2}$  and  $e_Q$  are defined in (2)–(5). The  $W_{0v}$  row was not computed in the archived runs.

| Geometry          | Method   | $e_u^{L^2}$ | $e_u^{H^1_{\text{br}}}$ | $e_p^{L^2}$ | $e_Q$  |
|-------------------|----------|-------------|-------------------------|-------------|--------|
| Uniform-27        | $W_{0n}$ | 0.2108      | 0.4483                  | 0.0321      | 0.1646 |
| Uniform-27        | $W_{1n}$ | 0.0816      | 0.2385                  | 0.0185      | 0.0243 |
| Uniform-27        | $W_{1v}$ | 0.0563      | 0.2161                  | 0.0217      | 0.0032 |
| Random-27         | $W_{0n}$ | 0.6532      | 0.8585                  | 0.1030      | 0.7272 |
| Random-27         | $W_{1n}$ | 0.3075      | 0.3897                  | 0.0322      | 0.1465 |
| Random-27         | $W_{1v}$ | 0.0651      | 0.1756                  | 0.0203      | 0.0271 |
| Heterogeneous-100 | $W_{0n}$ | 0.9254      | 1.0627                  | 0.0776      | 1.1199 |
| Heterogeneous-100 | $W_{1n}$ | 0.4738      | 0.5129                  | 0.0299      | 0.2595 |
| Heterogeneous-100 | $W_{1v}$ | 0.1998      | 0.3594                  | 0.0236      | 0.1244 |

$\|\xi_{0v} - \xi_{0n}\|_{\mathbb{S}_{uu}}^2$ ,  $\Delta_N = \|\xi_{1n} - \xi_{0n}\|_{\mathbb{S}_{uu}}^2$ , and the identity

$$\|\xi_{\text{CDD}} - \xi_{0n}\|_{\mathbb{S}_{uu}}^2 = \|\xi_{\text{CDD}} - \xi_{1v}\|_{\mathbb{S}_{uu}}^2 + \|\xi_{1v} - \xi_{1n}\|_{\mathbb{S}_{uu}}^2 + \|\xi_{1n} - \xi_{0n}\|_{\mathbb{S}_{uu}}^2, \quad (12)$$

and its analogue along  $W_{0n} \rightarrow W_{0v} \rightarrow W_{1v}$ . These identities are exact for nested spaces in exact arithmetic and their numerical verification requires solving the *full* C-DD system (the broken-pressure Galerkin solution  $\xi_{\text{CDD}}$ ), which was not executed in the archived runs. This verification is a planned item (see §10.3). The nestedness of the implemented bases is exact by construction, while §7.1–§7.2 verify the full-trace Schur elimination and the algebraic residual/conservation properties of the reduced systems. They do not establish equivalence between a reduced operator and the monolithic FEM operator.

## 8 Controlled modal experiments

### 8.1 Absolute accuracy

Table 4 collects the complete accuracy results of all archived runs: 3 geometries  $\times$  3 computed spaces, four metrics each. It is an absolute-results table (not improvement ratios).

Three observations follow directly from the absolute numbers, all qualitative and confined to the tested configurations:

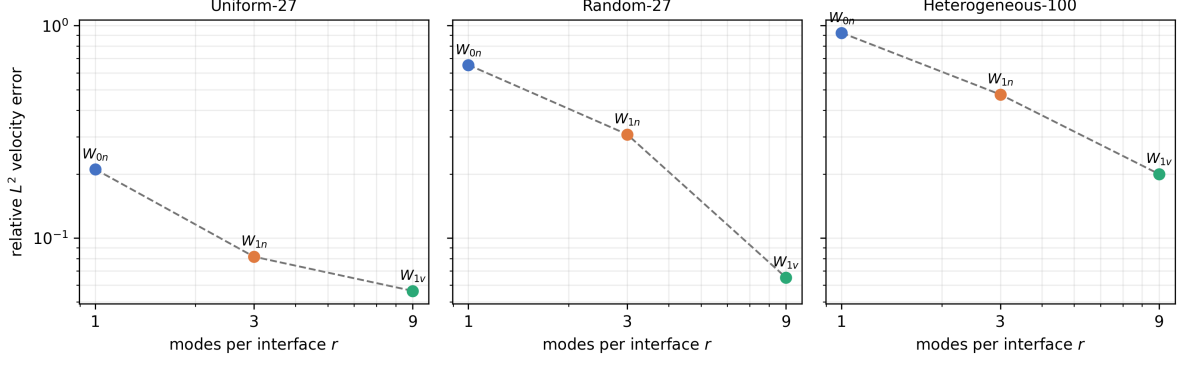


Figure 2: Relative  $L^2$  velocity error versus modes per interface ( $r = 1, 3, 9$ ) along the nested path  $\mathcal{W}_{0n} \rightarrow \mathcal{W}_{1n} \rightarrow \mathcal{W}_{1v}$ , for the three geometries (log scale). Only the path through  $\mathcal{W}_{1n}$  was computed; the equal-budget space  $\mathcal{W}_{0v}$  at  $r = 3$  is planned work.

- (C1). The Classic constant-normal model  $\mathcal{W}_{0n}$  exhibits large FEM-relative velocity errors on all three tested configurations (21.1%, 65.3% and 92.5%, respectively). These are observed model-to-FEM discrepancies; without the full C-DD reference they are not identified purely as modal-reduction errors.
- (C2). Along the computed path  $\mathcal{W}_{0n} \rightarrow \mathcal{W}_{1n} \rightarrow \mathcal{W}_{1v}$ , the first-order normal step accounts for the majority of the observed Classic-to-Affine  $L^2$  velocity-error reduction on all three configurations (quantified in §10.1).
- (C3). The additional full-vector affine enrichment remains substantial for Random-27 ( $e_u^{L^2}$ : 0.308  $\rightarrow$  0.065) and Heterogeneous-100 (0.474  $\rightarrow$  0.200), while it is a smaller field-error correction on Uniform-27 (0.082  $\rightarrow$  0.056). On Uniform-27 the outlet-flux error drops to 0.32% only with  $\mathcal{W}_{1v}$ .

The non-monotonicity of the pressure error on Uniform-27 ( $e_p^{L^2}(\mathcal{W}_{1n}) = 0.0185 < e_p^{L^2}(\mathcal{W}_{1v}) = 0.0217$ ) is not a discrepancy: the theory does not guarantee monotonicity of the pressure  $L^2$  error along nested chains (the reduction error is monotone in the Schur-energy seminorm, not in  $L^2(p)$ ).

## 8.2 Accuracy versus modal budget

Fig. 2 is the central visual of the section: for each geometry it plots the relative  $L^2$  velocity error against the number of modes per interface,  $r \in \{1, 3, 9\}$ , joining the points along the computed nested path  $\mathcal{W}_{0n} \rightarrow \mathcal{W}_{1n} \rightarrow \mathcal{W}_{1v}$ . The  $r = 3$  location carries a single computed space ( $\mathcal{W}_{1n}$ ); the equal-budget companion  $\mathcal{W}_{0v}$  is not yet computed, so the question “same DOF budget spent on direction versus on spatial variation” can only be posed here, not answered (see §8.4).

In absolute FEM-relative  $L^2$  error, the reduction from  $r = 1$  to  $r = 3$  is larger than the reduction from  $r = 3$  to  $r = 9$  in each of the three archived cases. This statement is intentionally additive rather than multiplicative: the log-scale ratios need not exhibit the same ordering. The remaining error at  $r = 9$  also differs substantially (5.6%, 6.5%, 20.0%), indicating configuration-dependent accuracy at a fixed modal budget; no asymptotic statement in  $r$  is claimed.

## 8.3 Metric comparison across geometries

Fig. 3 shows all four metrics as dot plots (velocity  $L^2$ , broken  $H^1$ , pressure  $L^2$ , outlet-flux error) with logarithmic ordinate; the wide dynamic range (0.3%–112%) is the reason for the log scale. The figure reproduces the Table 4 numbers in a form where the cross-geometry pattern is visible at a glance: the Classic-to-Affine spread widens from the regular to the irregular and heterogeneous cases in every metric except the pressure error, which stays within one decade across all geometries and methods.

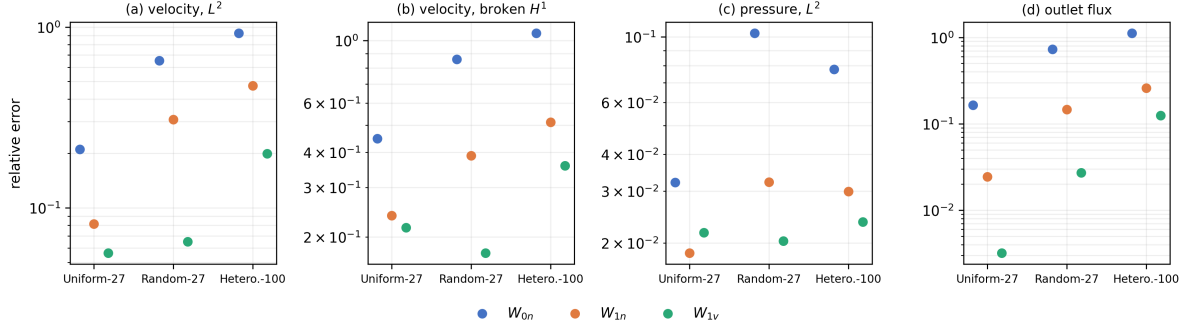


Figure 3: Four error metrics (relative, log scale) across the three geometries, three computed spaces: (a) velocity  $L^2$ , (b) velocity broken  $H^1$ , (c) pressure  $L^2$ , (d) outlet-flux error.

#### 8.4 Equal-budget comparison $\mathcal{W}_{0v}$ vs. $\mathcal{W}_{1n}$ : planned

The central controlled contrast of the programme is the equal-budget comparison

$$G_{0v} = \frac{e_{0n} - e_{0v}}{e_{0n}}, \quad G_{1n} = \frac{e_{0n} - e_{1n}}{e_{0n}}, \quad (13)$$

i.e. the FEM-relative  $L^2$  reduction obtained by spending three DOFs per interface on constant directional content ( $\mathcal{W}_{0v}$ ) versus on first-order spatial content ( $\mathcal{W}_{1n}$ ). This experiment was *not* run in the archived benchmarks (the constant-vector basis is implemented in the four-space framework but no runs were executed). It is the first item of §10.3. Until it is computed, the statement that the normal-linear modes are the dominant mechanism rests on the comparison  $\mathcal{W}_{0n} \rightarrow \mathcal{W}_{1n} \rightarrow \mathcal{W}_{1v}$  alone (Fig. 2), which cannot separate directional from spatial information.

#### 8.5 Field visualization

Fig. 4 shows, for Uniform-27 and Random-27, the mid-plane slice  $z = 0.5$  of the FEM velocity magnitude together with the pointwise velocity-error magnitude of the three reduced spaces. Within each row the three error panels share one color scale, so the reduced spaces are directly comparable panel to panel; the reference panel has its own scale. Axes are the physical coordinates of the unit cube. Solid outlines mark the sections of the solid spheres, and grey lines the intersections of the partition interfaces with the slice plane. Along the sequence  $\mathcal{W}_{0n} \rightarrow \mathcal{W}_{1n} \rightarrow \mathcal{W}_{1v}$  the error becomes both weaker and more spatially localized. On Uniform-27 the normal-linear space  $\mathcal{W}_{1n}$  already recovers most of the structure, while on Random-27 the residual  $\mathcal{W}_{1n}$  error is still spread over extended regions and only the affine space  $\mathcal{W}_{1v}$  approaches the FEM field, consistent with the global metrics (Table 4).

The slice fields are available for the two controlled geometries; per-interface normal-flux fields and Heterogeneous-100 field outputs remain planned work (see §10.3).

### 9 Larger heterogeneous configuration (Heterogeneous-100)

#### 9.1 Accuracy

Heterogeneous-100 is the largest of the three synthetic configurations (32 611 cells, 166 841 mixed DOFs, porosity 0.856). Its accuracy rows in Table 4 repeat the qualitative pattern of the controlled geometries, at a higher level: Classic  $\mathcal{W}_{0n}$  has a 92.5% relative velocity  $L^2$  error and a broken- $H^1$  relative error above one ( $e_u^{H^1_{br}} = 1.063$ ); NormalLinear reduces the  $L^2$  error to 47.4% and the flux error to 25.9%; Affine reaches  $e_u^{L^2} = 20.0\%$ ,  $e_Q = 12.4\%$ . The pressure error is the least sensitive metric here as well (7.8%  $\rightarrow$  3.0%  $\rightarrow$  2.4%). These are the largest errors among the three tested



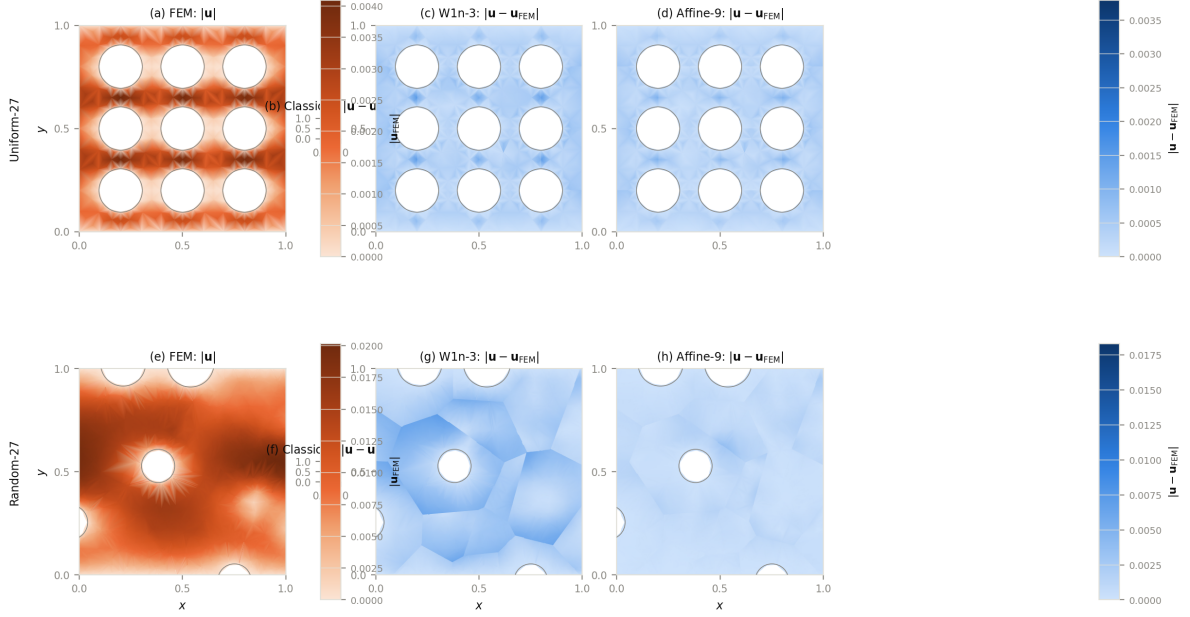


Figure 4: Slice plane  $z = 0.5$  of the velocity field and the FEM-relative velocity error on Uniform-27 (top row) and Random-27 (bottom row). Panels (a,e) show the FEM velocity magnitude  $|\mathbf{u}_{\text{FEM}}|$ ; panels (b,f), (c,g) and (d,h) show  $|\mathbf{u} - \mathbf{u}_{\text{FEM}}|$  for  $\mathcal{W}_{0n}$ ,  $\mathcal{W}_{1n}$  and  $\mathcal{W}_{1v}$ . Coordinates are physical (unit cube). Within each row the three error panels use a common color scale to enable direct visual comparison; the reference panel has its own scale. Solid outlines are the sections of the solid spheres and grey lines the partition-interface intersections with the slice plane.

configurations for each of the three computed spaces. The present data do not isolate a single geometric or topological quantity responsible for this ordering.

## 9.2 Cost and memory

Table 5 reports the cost structure of all runs. For Uniform-27 and Random-27 the reduced methods achieve a genuine first-solve speedup over the monolithic FEM ( $10.5\times$  and  $13.3\times$  for Classic;  $8.2\times$  and  $9.1\times$  for Affine), because the offline library is cheap relative to the FEM solve on those meshes. For Heterogeneous-100 the archive contains a 14.5 s FEM solve-only time, while the reduced offline phase takes 28–33 s. Thus the archived data do not support a like-for-like first-solve speedup claim for this case. The reduced online phase is short (0.06–1.4 s), and the quoted  $255\times$ ,  $72\times$  and  $10\times$  ratios compare that online phase against the archived FEM solve time only. A formal performance table should be rerun with one timing scope and repeated measurements on all configurations.

Memory: peak footprints on Heterogeneous-100 are 45 MiB (Classic), 79 MiB (NormalLinear), 246 MiB (Affine) versus 300 MiB for the monolithic solve. The observed increase with mode count is consistent with the larger reduced coupling blocks, but no memory-complexity law is inferred from these three measurements.

## 9.3 Accuracy–cost tradeoff

Fig. 5 plots the relative  $L^2$  velocity error of the three reduced spaces against their online time. The monolithic FEM solution has zero error relative to itself and therefore cannot be plotted on a logarithmic error axis; its archived time is shown only as a vertical reference line in each panel. The figure visualizes the reduced-method accuracy–online-cost tradeoff and is relevant primarily to repeated-query use, because it excludes the offline library. The Heterogeneous-100 FEM timing scope also differs from the other two cases, as stated in Table 5.



Table 5: Computational cost (seconds).  $T_{\text{offline}}$ : local factorisations + primitive responses;  $T_{\text{online}}$ : reduced assembly + Schur solve + reconstruction;  $T_{\text{first}} = T_{\text{offline}} + T_{\text{online}}$ .  $N_{\text{prim}} =$  primitive RHS columns. For Uniform-27 and Random-27 the FEM time is assembly+solve; for Heterogeneous-100 the archived FEM entry is solve-only, so timing ratios for that block are labelled “online only” and are not directly comparable to the first-solve ratios above.

| Geometry          | Method         | unknowns | $N_{\text{prim}}$ | $T_{\text{offline}}$ | $T_{\text{online}}$ | $T_{\text{first}}$ | speedup vs FEM            |
|-------------------|----------------|----------|-------------------|----------------------|---------------------|--------------------|---------------------------|
| Uniform-27        | $W_{0n}$       | 144      | 320               | 7.35                 | 0.0071              | 7.36               | $10.50\times$             |
| Uniform-27        | $W_{1n}$       | 432      | 896               | 7.38                 | 0.0222              | 7.40               | $10.44\times$             |
| Uniform-27        | $W_{1v}$       | 1296     | 2624              | 9.18                 | 0.208               | 9.39               | $8.23\times$              |
| Uniform-27        | FEM            | 63 955   | —                 | —                    | 77.25               | 77.25              | $1\times$                 |
| Random-27         | $W_{0n}$       | 114      | 246               | 8.90                 | 0.0141              | 8.91               | $13.29\times$             |
| Random-27         | $W_{1n}$       | 342      | 702               | 9.20                 | 0.0287              | 9.22               | $12.84\times$             |
| Random-27         | $W_{1v}$       | 1026     | 2070              | 12.82                | 0.191               | 13.01              | $9.10\times$              |
| Random-27         | FEM            | 75 954   | —                 | —                    | 118.42              | 118.42             | $1\times$                 |
| Random-27         | Exact-FE-Schur | 21 970   | —                 | —                    | 465.90              | 465.90             | $0.25\times$              |
| Heterogeneous-100 | $W_{0n}$       | 144      | —                 | 27.9                 | 0.057               | 27.98              | $255\times$ (online only) |
| Heterogeneous-100 | $W_{1n}$       | 432      | —                 | 28.0                 | 0.203               | 28.17              | $72\times$ (online only)  |
| Heterogeneous-100 | $W_{1v}$       | 1296     | —                 | 32.7                 | 1.400               | 34.13              | $10\times$ (online only)  |
| Heterogeneous-100 | FEM            | 166 841  | —                 | —                    | 14.53               | 14.53              | $1\times$                 |

Peak memory (Heterogeneous-100 only):  $W_{0n}$  45.0,  $W_{1n}$  78.8,  $W_{1v}$  246.4, FEM 299.8 MiB. “Online only”

speedups are  $T_{\text{online}}(\text{FEM})/T_{\text{online}}(X)$ ; these ratios exclude the reduced offline library and therefore do not constitute single-solve speedups.

Table 6: Cross-geometry modal summary: absolute errors and the observed FEM-relative reduction shares of (14).

| Geometry          | $e_{0n}^{L^2}$ | $e_{1n}^{L^2}$ | $e_{1v}^{L^2}$ | $R_N$ | $R_{V N}$ | reduction $\frac{e_{0n} - e_{1v}}{e_{0n}}$ |
|-------------------|----------------|----------------|----------------|-------|-----------|--------------------------------------------|
| Uniform-27        | 0.2108         | 0.0816         | 0.0563         | 83.6% | 16.4%     | 73.3%                                      |
| Random-27         | 0.6532         | 0.3075         | 0.0651         | 58.8% | 41.2%     | 90.0%                                      |
| Heterogeneous-100 | 0.9254         | 0.4738         | 0.1998         | 62.2% | 37.8%     | 78.4%                                      |

## 10 Cross-geometry modal analysis

### 10.1 Observed FEM-relative error-reduction shares

Along the computed path  $\mathcal{W}_{0n} \rightarrow \mathcal{W}_{1n} \rightarrow \mathcal{W}_{1v}$  the observed FEM-relative  $L^2$  velocity-error reduction decomposes into the normal-spatial step ( $\mathcal{W}_{0n} \rightarrow \mathcal{W}_{1n}$ ) and the additional vectorial step ( $\mathcal{W}_{1n} \rightarrow \mathcal{W}_{1v}$ ):

$$R_N = \frac{e_{0n} - e_{1n}}{e_{0n} - e_{1v}}, \quad R_{V|N} = \frac{e_{1n} - e_{1v}}{e_{0n} - e_{1v}}, \quad R_N + R_{V|N} = 1. \quad (14)$$

Fig. 6 and Table 6 report these shares.

Two statements can be made from these numbers, both confined to the tested configurations and to the chosen path:

- (D1). On all three geometries the normal-spatial step  $\mathcal{W}_{0n} \rightarrow \mathcal{W}_{1n}$  accounts for the majority of the observed reduction (59–84%).
- (D2). The share of the additional full-vector affine step  $\mathcal{W}_{1n} \rightarrow \mathcal{W}_{1v}$  is larger for Random-27 and Heterogeneous-100 (41% and 38%) than for Uniform-27 (16%). The simple partition-level variability descriptors in Table 1 do *not* order these three cases monotonically—Heterogeneous-100 has smaller listed CVs than Uniform-27—so the present data support configuration-dependent modal behavior but not a scalar “irregularity  $\mapsto$  modal share” law.

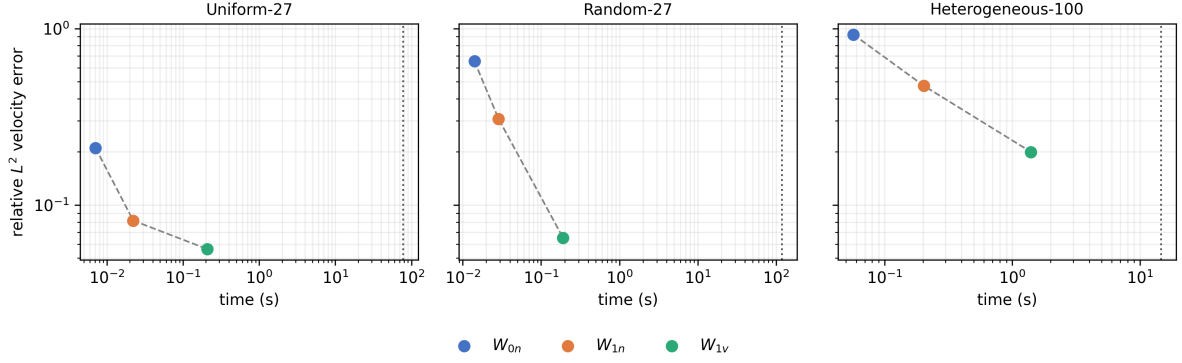


Figure 5: Relative  $L^2$  velocity error versus online solve time (log-log), shown separately for the three configurations. The vertical dotted line is the archived monolithic FEM time; the FEM self-error is exactly zero and is therefore off the logarithmic ordinate. Online time excludes the reduced offline library (see Table 5).

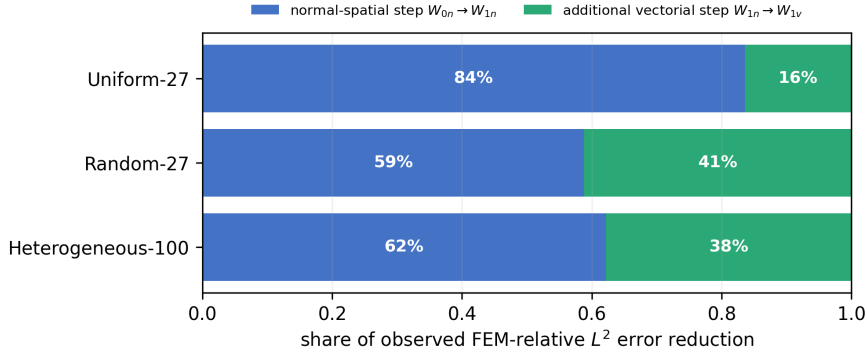


Figure 6: Fractions of the *observed* FEM-relative  $L^2$  velocity error reduction along the prescribed path  $W_{0n} \rightarrow W_{1n} \rightarrow W_{1v}$  (100% stacked). These fractions are descriptive; they are not unique causal modal contributions, because the two middle spaces  $W_{0v}$  and  $W_{1n}$  are generally non-nested and the path choice fixes the attribution.

## 10.2 Schur-energy pathwise decomposition: planned

The theory sections give an *exact* decomposition of the squared Schur-energy error into pathwise marginal gains (see §7.3). Unlike the FEM-relative shares of Fig. 6, those quantities ( $\Delta_N$ ,  $\Delta_{T|N}$  etc. in the  $\hat{\mathbf{S}}^{uu}$ -seminorm) are theorem-level and require the full C-DD trace solution  $\xi_{\text{CDD}}$ . They were not computed in the archived runs and are therefore not reported numerically; Fig. 6 and the planned Schur-energy figure must not be conflated. The former is an observed descriptive attribution, the latter would be a strict theorem quantity; only the former is available at this stage.

## 10.3 Geometry dependence and limitations

**What the current data supports.** (i) the Classic constant-normal model  $W_{0n}$  exhibits large FEM-relative velocity errors in all three tested configurations; (ii) adding first-order normal spatial content  $W_{1n}$  produces a major observed correction in each case; (iii) the residual gap between  $W_{1n}$  and  $W_{1v}$  is small for Uniform-27 and substantial for Random-27 and Heterogeneous-100, indicating configuration-dependent modal requirements across the tested cases; (iv) neither a monotonic dependence on a single scalar geometry descriptor nor a causal geometric law is claimed.

**What is planned and not yet computed.** (a) the equal-budget space  $W_{0v}$  (constant

tangential modes) on all three geometries, enabling the controlled contrast of §8.4; (b) the full C-DD solve  $\xi_{\text{CDD}}$  on Uniform-27 and Random-27 for the exact Schur-energy verification (§7.3) and the pathwise decomposition (§10.2); (c) a mesh-sensitivity study (Uniform-27 at three mesh levels) to confirm that the separation  $\mathcal{W}_{0n} \gg \mathcal{W}_{1n} > \mathcal{W}_{1v}$  is not an artifact of one mesh; the theory gives fixed-mesh separation only, no mesh-independent asymptotic statement; (d) an ensemble of random realisations of the Random-27 geometry to test the stability of the 59/41 split across seeds; (e) additional field outputs for Heterogeneous-100 and per-interface normal-flux fields ( $q = \mathbf{u} \cdot \mathbf{n}$ ) for the interface-structure visualization; (f) an independent second porous configuration at a different pore/sphere count; and (g) a uniform, repeated timing benchmark with the same FEM/reduced timing scope on every configuration.

**Scope notes.** All timings are single-machine, single-threaded measurements from the archived runs; they are indicative, not benchmark-suite grade. The reported errors are with respect to the monolithic Taylor–Hood FEM on the same mesh; the relation between the broken-pressure C-DD discretisation and the monolithic system is not established in the theory sections, and therefore no claim about the accuracy of the reduced methods *relative to the exact Stokes solution* is made.

## Appendix A: archived output provenance

All numerical values in this working section were read from the archived benchmark outputs listed below:

- `affine_ddpnm_3d/outputs/benchmark_w1n/`:  
`affine_ddpnm_metrics.csv`, `affine_ddpnm_report.json`, `affine_benchmark_fields.npz`;
- `affine_ddpnm_3d_random_porous/outputs/benchmark_w1n/`:  
`random_affine_metrics.csv`, `random_affine_report.json`, `random_benchmark_fields.npz`;
- `real_porous_benchmark_3d/outputs/benchmark_w1n/`:  
`benchmark_metrics.csv`, `benchmark_report.json`;
- the base-project verification run  
`ddpnm_3d_uniform_spheres/outputs/verify_run/` (Classic Schur eigenvalue footnote).

The geometry descriptors and tabulated numerical values are retained from the supplied working experiment document. In this revised version, Figs. 2, 3, 5 and 6 were regenerated directly from the tabulated values; Fig. 4 was regenerated from the field archives `affine_benchmark_fields.npz` and `random_benchmark_fields.npz`. No unreported experiment was assigned a numerical value.